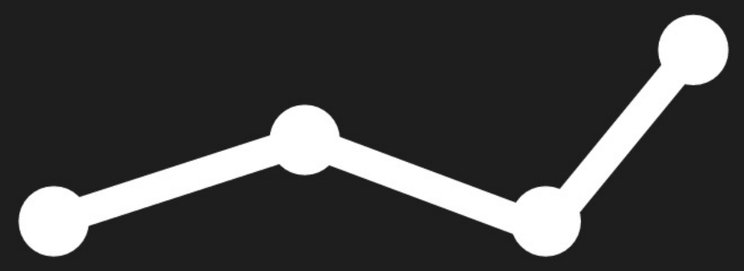
Your **6-Month** Learning Journey



Month 1

GenAI & Prompting Basics

Understand AI vs ML vs DL, GenAI platforms, ethics

Learn prompt anatomy & techniques (zero-shot, few-shot, ReAct, role-based)

Tools: Python, HuggingFace

Month 2

APIs & Real-World Prompting

Use LLM APIs (chat, embeddings)

Handle responses, manage API keys, debug outputs

Tools: LLM API

Month 3

NLP & Intro to RAG

Learn embeddings, tokenization, chunking, semantic search

Build basic RAG pipeline using LangChain/LlamaIndex

Tools: HuggingFace, LangChain/ LlamaIndex

Month 4

Advanced RAG

Multi-hop retrieval, hallucination reduction, fallback strategies

RAG evaluation & optimization

Tools: LangChain/LlamaIndex

Month 5

AI Agents & Multi-Agent Systems

Design single/multi-agent systems (ReAct, Tool-based, MARL)

Plan, reason & coordinate using LangChain, CrewAI/AutoGen

Tools: LangChain, LangGraph, CrewAI/ AutoGen

Month 6

Deploy & Orchestrate

Build & deploy APIs (FastAPI), frontend (Gradio/Streamlit)

Combine RAG + Agents + UI +monitoring (LLMOps)

Tools: FastAPI,, HuggingFace Spaces, Containerization

Detailed Curriculum
For your success

Module	Topics
MODULE 1 Prompt Engineering and API Integration	• Understanding AI vs ML vs DL • Introduction to Generative AI (GenAI) • Open-Source vs Closed-Source Models • GenAI Platforms Overview • Part 1: OpenAI, Anthropic, Mistral, Claude • Part 2: Stable Diffusion, Midjourney • Ethical Considerations in GenAI • Prompt Design Essentials • Anatomy of a Prompt (Role, Instruction, Input, Output) • Prompting Techniques: Zero-shot, Few-shot, Self-Consistency • Role-Based Prompting, ReAct (Reasoning & Acting) • API Integration Basics • Handling Responses (Parsing JSON, Error Handling) • Using OpenAI/Gemini APIs (chat, embeddings, function calling) • Rate Limits, Security, API Key Management • Debugging GenAI Outputs • Prompt Evaluation (Accuracy, Relevance)

MODULE 2

Basics of NLP and Retrieval-Augmented Generation (RAG)

- Text Preprocessing
  - Tokenization, Representation
  - Named Entity Recognition (NER), POS Tagging
- Embeddings & Semantic Search
  - Intro to Embeddings & Vectorization
  - Feature Engineering (Real-world Example)
- RAG Fundamentals
  - Why RAG? Benefits with LLMs
  - Core Components: Retrieval + Generation
  - Chunking Strategies, Data Handling, Indexing
- Vector Databases
  - Storing Text & Images
  - Vector Store Types, Search Strategies
- Building with RAG
  - Simple RAG Pipeline
  - Tools: LlamaIndex
  - Building a RAG App
- Advanced RAG Techniques
  - Multi-Hop Retrieval, Query Expansion
  - Reducing Hallucinations, Context Compression
  - Advanced Techniques: Self-Retrieval, LOTR, Multi-Query
- RAG Evaluation & Fallback Strategies

MODULE 3

AI Agents and Multi-Agent Systems

- Introduction to AI Agents
- Types of Agents: Reflex, Goal-Based, Utility-Based
- Agent Design Patterns
  - Reactive Agents, Tool-based Agents
  - Reference: AI Agent Patterns
- Multi-Agent Systems (MAS)
  - Need for MAS, Real-World Applications
  - Coordination: Negotiation, Consensus, Cooperation
- Popular Frameworks
  - ReAct, CrewAI, LangChain, LlamaIndex, Autogen
  - Comparative Use-Case Table
- Hands-On Use Cases
  - LangChain: ReAct Agent
  - LlamaIndex: Tool Calling / Compiler Agent
- Planning & Reasoning
  - Tool Integration & Orchestration
  - Hierarchical & Collaborative Agents
  - Multi-Agent Reinforcement Learning (MARL)
- Goal-Oriented Systems Design
- Latency & Cost Optimization

MODULE 4

Deployment and Project Integration

- Web API Basics (GET, POST, PUT, DELETE)
- Building APIs with FastAPI (Swagger, Pydantic)
- Deploying APIs & Apps
  - Platforms: Render, Railway, Replit, Docker, HF Spaces
- CI/CD with GitHub Actions
- GenAI Architecture Recap
  - Agents, RAG, API Orchestration
- Designing Modular Systems (RAG, Memory, UI)
- FastAPI for LLM Orchestration
- Frontend Integration (Gradio, Streamlit)
- Security & Monitoring
  - API Key Management, Rate Limiting
  - Error Handling & Monitoring
- LLMOps Overview
  - Need for LLMOps, Prompt Versioning, Tracing
  - Platforms: Langfuse, LangGraph, MLflow
- Live Deployment Demo & Project Showcase

MODULE 5
(OPTIONAL)

Foundations of Neural Networks and Transformers

- Linear & Logistic Regression
- Perceptron and Multi-Layer Perceptron (MLP)
- Activation & Loss Functions
- Backpropagation & Gradient Descent
- Deep Learning: CNNs, RNNs, LSTMs
- Evolution to Transformers
- Attention Mechanism & Transformer Architecture
- Pretraining vs Finetuning
- Transfer Learning

MODULE 6
(OPTIONAL)

Fine-Tuning LLMs

- When to Fine-Tune vs. Use RAG
- Choosing Models: LLaMA, Mistral, Phi
- Dataset Prep: Alpaca, DPO
  - Cleaning, Labeling, Formatting
- PEFT Techniques
  - LoRA, QLoRA
  - Training with HuggingFace, PEFT, Datasets
- Inference & Quantization
- Evaluation & Costing
  - Metrics: Accuracy, F1, BLEU, ROUGE
  - Comparing before vs. after fine-tuning
  - Cost Estimation & Performance Trade-offs

Projects You Will Build

Module 1: Prompt Engineering and API Integration

1. AI-Powered Scam Detection System

Tackle spammy content in emails and SMS texts by building an advanced AI scam detector that restricts fraudulent attempts - helping customers avoid the possible risk by explaining them in layman's terms.

Skills Mastered: LLMs for Message Analysis, AI Text Classification, Intent Extraction, Fraud Pattern Detection.

2. AI-Powered Product Insight Generator

Turn messy customer reviews into sorted product insights by building an intelligent insight generator that analyzes reviews and generates summaries - helping customers make informed decisions.

Skills Mastered: Prompt Chaining, Structured Output Generation, Dynamic Summarization Techniques.

Module 2: Basics of NLP and Retrieval-Augmented Generation (RAG)

1. Resume-to-Job Semantic Search Engine

Build an advanced AI matching engine that functions on contexts and enables HRs to find relevant candidates for a job opening, even if the candidates don't have exact phrases in their resumes.

Skills Mastered: Embeddings, Vector Databases, MLOps & Privacy.

2. Smart Helpdesk Assistant

Build an intelligent AI chatbot that optimizes customer support for an organization by responding in the most logical and accurate manner to real-time queries - using internal data and documents of an organization.

Skills Mastered: Retrieval-Augmented Generation (RAG), Retrieval Pipelines, Prompt Engineering.

Module 3: AI Agents and Multi-Agent Systems

1. AI-Powered Finance and Inventory Management Assistant

Optimize financial operations and inventory control process of a business with the help of an advanced AI assistant - based on data-driven insights that enable businesses to make more informed decisions.

Skills Mastered: Agent-Based Workflows, Structured Memory, Reasoning Chains.

2. Custom Agentic AI Ecosystem

Enter the role of a GenAI architect, where you're no longer just building a single model - you're designing an entire ecosystem of intelligent agents that work collaboratively to solve dynamic, real-world problems.

Skills Mastered: Agentic AI, Workflow Optimization & Automation, Ecosystem Integration.

Module 4: Deployment and Project Integration

1. AI-Powered HealthCare Management System

Transform fragmented patient experiences into seamless healthcare journeys by creating an effective AI assistant that understands symptoms, recommends the right doctors, automates appointment scheduling, and generates contextual summaries.

Skills Mastered: Multi-Agent Coordination, Retrieval-Augmented Generation (RAG).

2. Smart Customer Support Chatbot System

Build a full-stack GenAI chatbot that can power enterprise customer support by retrieving real-time answers from documentation, handling complex queries, and triggering workflows - for B2B and B2C as well.

Skills Mastered: Prompting, RAG, Multi-Agent Systems, Deployment.

By the end of the program, you will be able to-

Write AI prompts

Skills: Prompt Engineering, AI vs ML vs DL, Open-source vs. closed-source models

What you'll be able to do: Learn how to craft precise prompts to get the best responses from AI. Guide AI to act as a specific role and generate relevant outputs for your projects.

Tools: Python, HuggingFace

Integrate GenAI into Projects

Skills: API Integration, Prompt Evaluation, Security, Debugging

What you'll be able to do: Seamlessly integrate GenAI into your apps and evaluate performance. Identify and fix errors while ensuring your implementation is secure and scalable.

Tools: HuggingFace API, OpenAI API / Gemini API

Make Smarter AI Assistants

Skills: NLP & Retrieval-Augmented Generation (RAG)

What you'll be able to do: Build smarter GenAI chatbots, legal assistants, and research tools that can retrieve and use real-time external data during live interactions.

Make Smarter AI Assistants

Skills: NLP & Retrieval-Augmented Generation (RAG)

What you'll be able to do: Build smarter GenAI chatbots, legal assistants, and research tools that can retrieve and use real-time external data during live interactions.

Tools: LangChain, LlamaIndex, HuggingFace Embeddings

Humanize Your AI Assistants

Skills: AI Agents, Multi-Agent Systems

What you'll be able to do: Build multi-agent AI systems where each agent takes on a specialized role, plans tasks, and collaborates to achieve complex goals autonomously.

Tools: LangChain, LangGraph, AutoGen/CrewAI, HuggingFace Transformers

Deploy AI into the Real World

Skills: AI Deployment and Project Integration

What you'll be able to do: Launch your GenAI apps or assistants with secure API access using scalable, production-ready deployment practices.

Tools: FastAPI, Containerization tools

Know How GenAI Functions

Skills: Neural Networks and Transformers

What you'll be able to do: Understand the inner workings of GenAI tools like ChatGPT and Gemini, including the neural network and transformer architectures that power them. Implement AI programs more effectively with this deeper knowledge.

Tools: Pytorch/TensorFlow/Keras, HuggingFace Transformers, BertViz

Build Your Own ChatGPT

Skills: Fine-Tuning LLMs

What you'll be able to do: Take your GenAI skills to the expert level by fine-tuning LLMs to build your own GPT-style assistants for enterprise, research, or custom use cases.

Tools: HuggingFace Trainer, Transformers

Career Paths
You Can Pursue

After this program, you won't just know GenAI, you'll be able to build with it.

Here's a peek at the roles you can confidently explore:

Applied AI & GenAI Roles

- Generative AI Engineer
- AI Research Engineer
- Prompt Engineer
- LLM Application Developer
- NLP Engineer (LLMs & Transformers)
- AI Product Developer

AI Strategy & Consulting

- AI Solutions Architect
- GenAI Consultant
- AI/ML Product Manager

Software & Research Role

- Full Stack AI Developer (with GenAI capabilities)
- Research Assistant (in AI/ML domains)
- Open Source AI Contributor (e.g., LangChain, HuggingFace)

GEN AI

Cross-Functional & Emerging Roles

- Conversational AI Specialist (Chatbots, Voice Agents)
- Human-AI Interaction Designer
- AI Content Engineer (e.g., for marketing, education)

Applied Multi-Agent Systems (MAS) RolesAI & GenAI Roles

- Multi-Agent Systems Engineer
- Autonomous Systems Developer
- AI Systems Architect
- Simulation Engineer

Tools & **Technologies**

Python

HuggingFace

OpenAI API / Gemini API

LangChain/
LlamaIndex

HuggingFace
Embeddings

LangGraph
(Framework)

AutoGen
/CrewAI

HuggingFace
Transformers

FastAPI

BertViz

HuggingFace
Trainers

PyTorch/TensorFlow
/Keras

**Containerization Tools
(Deployment Tools)**